

Procedure and the Power Approach for Assessing the Equivalence of Average Bioavailability

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Journal of Pharmacokinetics and Biopharmaceutics, Vol. 15, No. 6, pp. 657-680, 1987 | DOI:
10.1007/BF01068419

Methodology Overview

Schuirmann introduced the Two One-Sided Tests (TOST) procedure to resolve the logical contradiction of using standard hypothesis testing (testing for difference) to claim equivalence. TOST decomposes equivalence into two simultaneous composite null hypotheses: (1) difference is less than the lower equivalence bound $-\delta$, and (2) difference is greater than the upper equivalence bound $+\delta$. Equivalence is declared at level α if and only if both one-sided nulls are rejected, which is mathematically equivalent to the $(1 - 2\alpha)$ confidence interval falling entirely within $(-\delta, +\delta)$.

Role in Our VLM-DB Statistical Verification

Employed in Section 4.1 to formally verify statistical equivalence between control configurations and establish that lack of difference is a genuine invariant rather than insufficient statistical power.